**R16**

**Code No: 136AQ**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year II Semester Examinations, May - 2019**
**COMPILER DESIGN**
**(Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART - A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define regular expression. | [2] |
| b) | Define linker and loader and explain briefly. | [3] |
| c) | Define ambiguous grammar. | [2] |
| d) | Compare SLR, CLR and LACR. | [3] |
| e) | What is coercion? | [2] |
| f) | How to find evaluation order for SDD's? | [3] |
| g) | What are the limitations of static allocation? | [2] |
| h) | Write the fields and uses of symbol table. | [3] |
| i) | What is common sub-expression elimination? Explain. | [2] |
| j) | What are induction variables? What is induction variable elimination? | [3] |

**PART - B**

**(50 Marks)**

2.a) Explain the procedure to convert regular expression to Finite automata.
b) Explain various phases in the construction of compiler with a neat sketch. [5+5]

**OR**

3.a) What is the functionality of preprocessing and input buffering?
b) Explain compiler construction tools. [5+5]

4.a) What is left recursion? Describe the algorithm used for eliminating left recursion?
b) Eliminate left recursion in the following: [5+5]
E→E+T|T, T→T*F|F, F→(E)|id

**OR**

5.a) What is ambiguous grammar? Show that following grammar is ambiguous or not.
A → A + A | A − A | A * A | a
b) Verify whether the following grammar is LL(1) or not? [5+5]
E → E + T | T
T → T* F / F
F → (F) |a|b.

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6.a) What are three address codes? Explain different types of representations of three address code.
b) Write three codes for x:=A[y, z] [5+5]

**OR**

7.a) What is type checker? Explain the specification of a simple type checker.
b) Explain translation schema for array elements. [5+5]

8.a) Explain about Heap management.
b) Define reference counting. What is the role of reference counting in garbage collection? [5+5]

**OR**

9.a) Give the detailed description on DAG.
b) Explain different methods for register allocation and assignment. [5+5]

10.a) Explain redundancy elimination techniques.
b) Write the principal sources of optimization. [5+5]

**OR**

11.a) Explain loop optimization technique with example.
b) Explain constant propagation with example. [5+5]

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